Velocity and position graphs

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Connect a position-time graph to velocity using slope.

SUCCESS CRITERIA - I can...

1. Read positions and elapsed times from a graph.
2. Find velocity from position change divided by time.
3. Explain the difference between speed and velocity.

Core: teacher selects from 1-9. **Stretch:** 10 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “velocity” and “speed” to explain your model or result. Sentence starters are optional.

velocity: rate of position change, with direction

speed: how fast something moves, without direction

slope: vertical change divided by horizontal change on a graph

01 Warm-up: bring back an earlier idea

East is positive. Start at 5 m, go to -3 m, then finish at 2 m.
Sketch the route; calculate distance and displacement. Distinguish both from final position.

Word help: displacement = change from initial position to final position, with direction

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

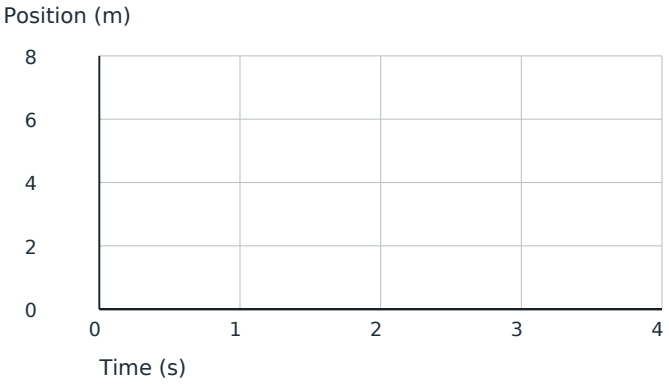
02 Worked example: explain the model

Reference: $v_{avg} = \Delta x/\Delta t$; average speed = distance/Δt. Δ means change; Δt is elapsed time. Position-time slope gives velocity.

Worked reasoning: A cart moves from 1 m to 7 m in 3 s at steady velocity. The position-time slope is (7-1)/3=2 m/s.

Your turn: Use the blank axes to plot the cart's two measured positions and connect them.
Draw a slope triangle.
Label its rise in meters and its run in seconds.

Sentence starter: The ... in my model represents ... . Its label shows ... .



Marzano check - circle: ① ② ③

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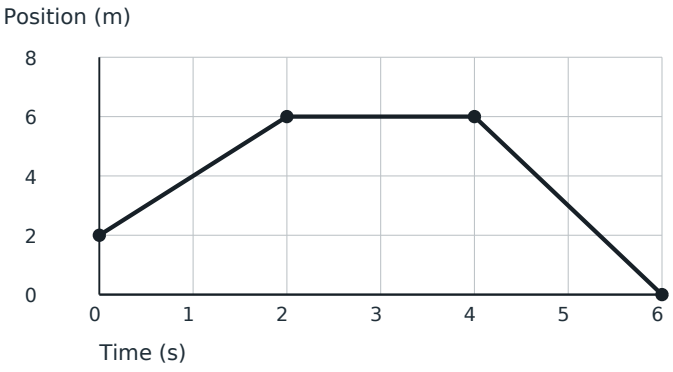
Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Interpret all three intervals

For the displayed position-time graph, calculate velocity on each segment.
Label the direction of motion on each.

Sentence starter: I found ... [unit]. This value means ... in this situation.



Marzano check - circle: ① ② ③

04 Compare two averages

For the full 6 s trip in 3, calculate average speed and average velocity.
Explain their difference using the route.

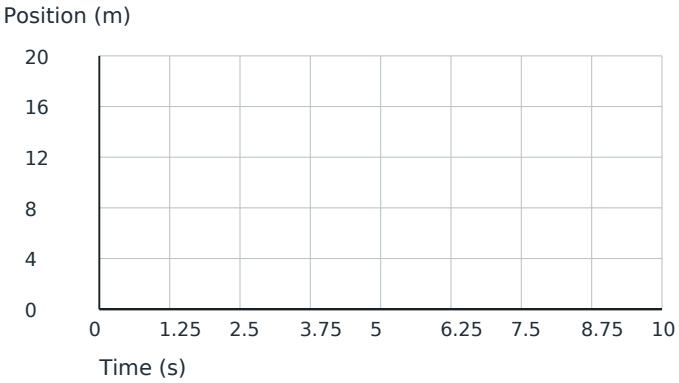
Sentence starter: Both ... . They differ in ... because ... .

Marzano check - circle: ① ② ③

05 A line that misses the origin

A robot's position-time graph passes through (2 s, 14 m) and (6 s, 6 m). It moves at constant velocity. Plot the line, calculate its slope, and write x(t).
Predict its position at 9 s.

Sentence starter: I predict ... . My model supports this because ... .

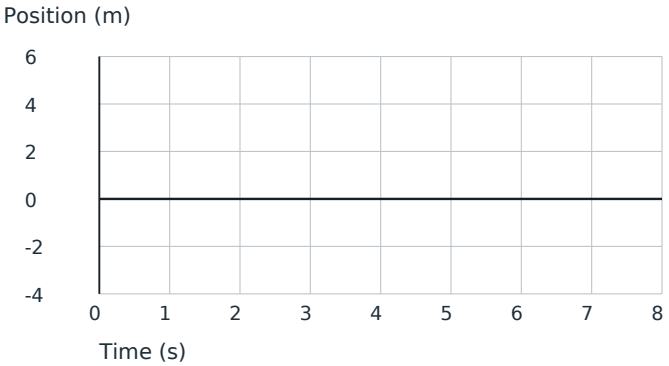


Marzano check - circle: ① ② ③

06 Construct the graph

A robot starts at -2 m. It moves at +3 m/s for 2 s, waits for 2 s, then moves at -2 m/s for 3 s.
Draw its complete position-time graph and label the three intervals.

Sentence starter: The ... in my model represents ... . Its label shows ... .



Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

07 Separate height and slope

A student reads a point at (5 s, 10 m) and reports a velocity of 2 m/s. Under what additional condition would that calculation be justified?

Sentence starter: I notice … . This supports … because … .

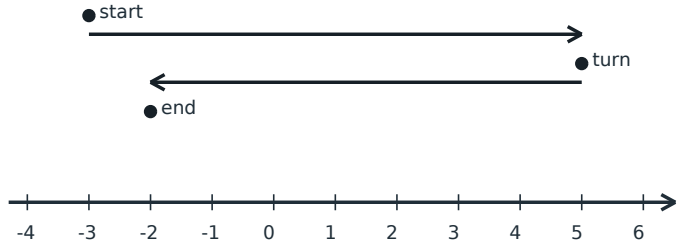
Marzano check - circle: 1 2 3

08 Review: connect an earlier idea

East is positive. Start at -3 m, go to 5 m, then finish at -2 m.
Sketch the route; calculate distance and displacement. Distinguish both from final position.

Word help: displacement = change from initial position to final position, with direction

Sentence starter: I found … [unit]. This value means … in this situation.



Position (m); right is positive

Marzano check - circle: 1 2 3

09 Exit check

A straight segment runs from (1 s, -2 m) to (5 s, +6 m).
Find velocity and state what the slope units mean.
Predict x at 7 s if the same motion continues.

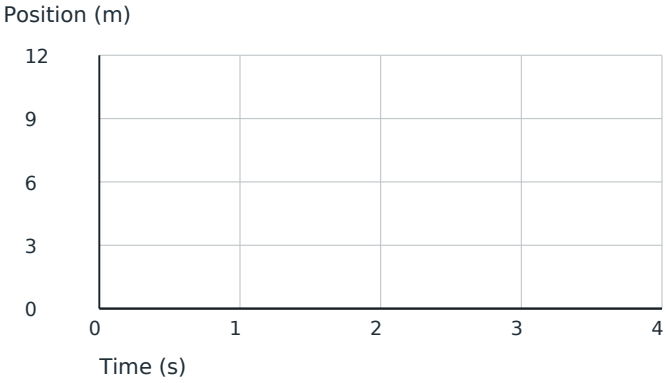
Sentence starter: I predict … . My model supports this because … .

Marzano check - circle: 1 2 3

10 STRETCH: test the model

Two carts follow $x_A = 1 + 2\,t$ and $x_B = 9 - t$, in meters and seconds.
Sketch both graphs and find when and where they meet.

Sentence starter: I found … [unit]. This value means … in this situation.



Marzano check - circle: 1 2 3